Sahaj Bhadja

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Summary

Platform engineer with 2+ years building **Python web applications**, tuned **SQL/ETL workflows**, **LLM integrations**, and Docker/CI/CD releases; delivered systems supporting 10K+ daily requests and 99%+ reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python FastAPI platform services** and normalized SQL schemas for 500+ users, tuning indexed queries and access patterns to reduce daily dashboard latency from **1.2 seconds to 350 milliseconds**.
- Implemented **data-validation, reconciliation, and error-handling workflows** across 5,000+ monthly ETL jobs, achieving 99%+ production reliability while accelerating diagnosis of malformed records, failed integrations, and recurring runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Delivered **full-stack React and Flask features** for 150+ instructors, translating upload, grading-history, and reporting requests into maintainable interfaces and APIs handling 10,000+ daily grading requests in production.
- Integrated **multimodal LLM and OCR providers** behind switchable APIs, adding usage logging, fallback handling, and concurrent processing to reduce feedback turnaround 60% while improving production supportability for instructors.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python data and evaluation pipelines** processing 100,000+ trajectory records, creating reproducible test scenarios and diagnostics that improved long-horizon planning 18% and strengthened investigation of model behavior.
- Documented **model constraints, benchmark results, and failure patterns** in technical reports, translating complex experimental behavior into actionable guidance that improved closed-loop planning stability by 15% across simulations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Optimized **large-scale API and ETL integrations** using Python, AWS Glue, S3, SQL, and caching, improving response times 30% and pipeline efficiency 68% across distributed data workflows in production.
- Automated **cross-platform builds, deployments, and 20+ unit/integration suites** with Jenkins, PyTest, and Mockito, reducing regressions and technical debt 20% while supporting reliable releases across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **React, Node.js, and Azure SQL features** across eight internal CRM workflows, synchronizing 2M+ records with WebSockets while improving usability for resource, billing, and permissions operations daily.
- Packaged services with **Docker and automated Jenkins CI/CD**, testing, staged deployments, and monitoring, cutting release time 50% while reducing environment drift across fast-paced development and support cycles.

Skills

Languages: **Python**, **SQL**, JavaScript, TypeScript, Java, C++, Bash, HTML, CSS
Web & APIs: React.js, Next.js, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, joins, aggregations, query tuning, indexing, data validation
AI & Testing: LLM APIs, tool integrations, OCR, PyTorch, automated testing, PyTest, Jest, Selenium
Platform: Docker, Linux, AWS, Microsoft Azure, Jenkins, GitHub Actions, CI/CD, ETL pipelines, Git, JIRA

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL development workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript codebase with Docker-backed development workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **fintech web platform** supporting 50,000+ users and 1M+ transactions, combining Next.js filtering, Plaid bank linking, real-time updates, MongoDB access, and Kubernetes-based CI/CD deployments for resilient operations.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python data-processing API** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.